SOFTWARE TEST PLAN (STP)

Task Manager Application

Project: Task Manager Application
Team: Team 6
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Team Members

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Technology Stack

JavaScript (React.js + Node.js) / Python (Django)

1. Introduction

1.1 Purpose

The purpose of this Software Test Plan is to describe how the Task Manager Application will be tested. The application helps individual users and teams create, organize, assign, track, and complete tasks. The testing process will check whether the implemented system satisfies the functional, non-functional, and security requirements given in the Software Requirements Specification (SRS). This document explains the features to be tested, testing methods, test environment, security validation, test cases, risks, and requirement traceability.

1.2 Scope

Testing covers the main functions of the Task Manager Application, including:

- User registration and login
- Password management
- Task creation, editing, and deletion
- Task status and priority management
- Filtering, sorting, and searching tasks
- Notifications and reminders
- Task assignment and team collaboration
- Dashboard and reports
- Security and authorization
- Performance and responsiveness

Payment systems and third-party calendar synchronization are outside the scope of Version 1.0, as mentioned in the SRS.

1.3 References

- Task Manager Application SRS Version 1.0
- Software Architecture and Design Specification
- Software Engineering Mini-Project Deliverables
- IEEE Software Testing Documentation Guidelines

1.4 Definitions

Term	Meaning
SRS	Software Requirements Specification
STP	Software Test Plan
FR	Functional Requirement
NFR	Non-Functional Requirement
API	Application Programming Interface
REST	Representational State Transfer
JWT	JSON Web Token
CRUD	Create, Read, Update, Delete
RBAC	Role-Based Access Control
TLS	Transport Layer Security
RTM	Requirements Traceability Matrix

2. Test Items

1. Authentication and Account Management Module
2. Task Management Module
3. Task Organization Module
4. Notification Module
5. Collaboration Module
6. Reporting and Dashboard Module
7. REST API
8. Database interaction
9. Security mechanisms
10. User Interface

3. Features to be Tested

The following features are selected based on the requirements defined in the SRS.

3.1 Authentication

TM-F-001 – User Registration: The system should allow a new user to register using name, email, and password.

TM-F-002 – User Login: The system should authenticate a registered user using email and password before providing access.

TM-F-003 – Secure Password Storage: Passwords should be stored only after hashing and salting. Plaintext passwords should not be stored.

TM-F-004 – Password Reset: The application should allow the user to reset a forgotten password using a time-limited email link.

TM-F-005 – Account Lock: The account should be locked for 15 minutes after five consecutive failed login attempts.

3.2 Task Management

TM-F-010 – Create Task: The user should be able to create a task with a title, description, and due date.

TM-F-011 – Edit Task: The user should be able to edit an existing task.

TM-F-012 – Delete Task: The owner should be able to delete a task after confirmation.

TM-F-013 – Update Task Status: The user should be able to set a task as To-Do, In Progress, or Completed.

TM-F-014 – Set Priority: The user should be able to set Low, Medium, or High priority.

TM-F-015 – Tags/Categories: The user should be able to assign categories or tags to tasks.

3.3 Task Organization

TM-F-020 – Filter Tasks: Filter tasks by status, priority, category, and due date.

TM-F-021 – Sort Tasks: Sort tasks by due date, priority, or creation date.

TM-F-022 – Search Tasks: Search tasks using keywords in the title or description.

TM-F-023 – Projects/Boards: Organize tasks into projects or boards.

3.4 Notifications

TM-F-030 – Due-Date Notification: Notify the user when a task is due within 24 hours.

TM-F-031 – Overdue High-Priority Reminder: Send an email reminder for overdue high-priority tasks.

TM-F-032 – Assignment Notification: Notify a user in real time when a new task is assigned.

3.5 Collaboration

TM-F-040 – Invite Team Members: Allow a task owner to invite another registered user to a project.

TM-F-041 – Assign Tasks: Allow a user to assign a task to another team member.

TM-F-042 – Comments: Allow team members to add comments to shared tasks.

3.6 Reporting

TM-F-050 – Dashboard: Display task counts by status.

TM-F-051 – Weekly Productivity Report: Show tasks completed versus tasks created.

TM-F-052 – Admin Analytics: Allow an admin to view aggregate usage analytics.

3.7 Non-Functional Features

TM-NF-001 – Performance: Page load and API response time should be ≤ 2 seconds for 90% of requests under normal load of up to 500 concurrent users.

TM-NF-002 – Reliability: The application should provide 99.5% monthly uptime, excluding scheduled maintenance.

TM-NF-003 – Security: Data transmitted through the application should use TLS 1.2 or above, and protected APIs should reject unauthenticated requests.

TM-NF-004 – Usability: The interface should work on screen sizes from 360px mobile screens to 1920px desktop screens.

TM-NF-005 – Scalability: The backend should support scaling up to 5,000 concurrent users without changes to application code.

4. Features Not to be Tested

- Internal implementation of React.js
- Internal implementation of Node.js or Django
- Browser source code
- Operating system functionality
- Third-party email service internal logic
- OAuth provider internal implementation
- Payment systems
- Third-party calendar synchronization
- External cloud infrastructure not controlled by the team

Only the interaction of these external services with our application will be tested where required.

5. Test Approach / Strategy

Testing will be performed at different levels and will include functional, non-functional, integration, usability, regression, performance, and security testing.

5.1 Security Validation

Security testing is required because the application stores user account information and private task/project data.

Security Objectives

- SO-01 – Confidentiality: User credentials and task/project information should be protected from unauthorized users.

- SO-02 – Integrity: Task and project information should not be modified by unauthorized users or corrupted during transmission.
- SO-03 – Availability and Accountability: Only authenticated and authorized users should perform actions, and important actions should be traceable through audit logs.

Security Requirements

TM-SR-001 – TLS Security: TLS 1.2 or higher should be used for client-server and server-database communication.

TM-SR-002 – Password Security: Passwords should be stored using salted one-way hashing such as bcrypt or an equivalent method.

TM-SR-003 – Input Validation: API inputs should be validated and sanitized to reduce SQL injection and cross-site scripting attacks.

TM-SR-004 – Role-Based Access Control: A user should only access tasks and projects that they own or are authorized to access.

TM-SR-005 – Rate Limiting: Authentication endpoints should allow no more than 10 requests per minute per IP address. The eleventh request within the same minute should be rejected.

5.2 Unit Testing

Individual modules and functions will be tested separately. Examples include registration validation, login validation, task creation, task editing, task status updates, and search functionality.

5.3 Integration Testing

Integration testing will check communication between the React frontend and backend API, backend API and database, login module and dashboard, and task assignment and notification system.

5.4 System Testing

The complete application will be tested as a single system. Example end-to-end flow: Register → Login → Create Task → Assign Task → Update Status → View Dashboard.

5.5 User Acceptance Testing

The application will be tested from the user's point of view to check whether the major project requirements are satisfied.

5.6 Functional Testing

Functional testing will check whether the application performs the functions stated in the SRS.

5.7 Regression Testing

Previously tested functions will be tested again after changes or bug fixes.

5.8 Performance Testing

Response time and system behaviour under user load will be checked.

5.9 Usability Testing

The interface will be checked for easy navigation, readable text, clear buttons, responsive layout, and proper messages.

5.10 Entry and Exit Criteria

Entry Criteria

- The major application modules are implemented and available for testing.

- The test environment is properly configured.
- Required test data and test user accounts are available.
- The frontend, backend, and database can communicate successfully.
- The build is stable enough for test execution.

Exit Criteria

- All planned test cases have been executed.
- All high-priority functional requirements have been tested.
- No critical defects remain unresolved.
- Required security validation tests have been completed.
- Non-functional requirements included in the test plan have been tested.
- The Requirements Traceability Matrix has been updated with the final results.

6. Test Environment

Hardware

- Laptop or desktop computer
- Minimum 4 GB RAM
- Internet/network connection

Software

- Windows/macOS/Linux
- Google Chrome
- Firefox
- Safari or Edge
- React.js frontend
- Node.js or Django backend
- PostgreSQL/MySQL database
- Visual Studio Code

Testing Tools

- Browser Developer Tools
- Postman
- OWASP ZAP
- GitHub
- Jira
- Manual testing tools

Test Data

- Test user accounts
- Sample emails
- Sample passwords
- Task titles
- Descriptions
- Due dates
- Task priorities
- Categories
- Invalid credentials

- Invalid form input

7. Test Schedule

Activity	Planned Time
Requirement review	September 2026
Test plan preparation	September 2026
Test case preparation	September 2026
Environment setup	September 2026
Functional testing	September 2026
Integration testing	September 2026
Security testing	September 2026
Defect fixing and retesting	September 2026
Final review	Before submission

8. Test Deliverables

- Software Test Plan
- Test cases
- Test data
- Test execution results
- Defect reports
- Requirements Traceability Matrix
- Test Summary Report

9. Roles and Responsibilities

Role	Responsibility
Team Members	Development and testing
Test Designer	Prepare test cases
Tester	Execute test cases and record results
Developer	Fix reported defects
Team	Review final results

Since this is a student mini-project, responsibilities can be shared between all four team members.

10. Risks and Mitigation

Risk	Mitigation
Development delay	Complete high-priority modules first

Risk	Mitigation
Database connection issue	Verify configuration and maintain backup
Frontend/backend integration issue	Perform integration testing regularly
Bugs close to submission	Give priority to critical defects
Email service unavailable	Use test or mock email services
Limited testing time	Prepare test cases before implementation is complete
Loss of code	Use GitHub/version control

11. Assumptions and Dependencies

- Supported browsers are available.
- The database is configured correctly.
- The application server is available.
- Test accounts and test data are available.
- The email service required for password reset and notifications is available.
- The frontend and backend modules are sufficiently stable before system testing.
- Internet/network connectivity is available where required.

12. Suspension and Resumption Criteria

Suspension Criteria

- The application cannot start.
- Database connectivity fails.
- Login functionality is completely unavailable.
- A critical defect blocks most test cases.
- Required modules are not implemented.
- The test environment is unavailable.

Resumption Criteria

- Blocking defects are corrected.
- The database connection is restored.
- Application stability is restored.
- Required modules become available.
- The test environment is functioning normally.

13. Test Case Management and Traceability

The test cases are linked directly to the requirements defined in the SRS. The following 10 test cases cover important functional, non-functional, and security requirements.

Test Case 1 – User Registration

Test Case ID: TC-Auth-01

Requirement ID: TM-F-001

Objective: Verify that a new user can register successfully.

Precondition: The email address has not already been registered.

Test Steps:

1. Open the registration page.
2. Enter name.
3. Enter a valid email.
4. Enter a valid password.
5. Click Register.

Expected Result: A new user account should be created and confirmation should be sent.

Test Case 2 – User Login

Test Case ID: TC-Auth-02

Requirement ID: TM-F-002

Objective: Verify that a registered user can log in.

Precondition: A valid account already exists.

Test Steps:

1. Open the login page.
2. Enter valid email.
3. Enter correct password.
4. Click Login.

Expected Result: The user should be authenticated and provided access to the application.

Test Case 3 – Password Security

Test Case ID: TC-Auth-03 / TC-Sec-02

Requirement ID: TM-F-003, TM-SR-002

Objective: Verify that passwords are not stored in plaintext.

Precondition: A user account has been created.

Test Steps:

1. Register a test user.
2. Inspect the stored password value in the test database.
3. Check the application logs.

Expected Result: The password should appear only as a salted hash and should not appear in plaintext in the database or logs.

Test Case 4 – Create Task

Test Case ID: TC-Task-01

Requirement ID: TM-F-010

Objective: Verify that a logged-in user can create a new task.

Precondition: The user is logged in.

Test Steps:

1. Select Create Task.
2. Enter task title.
3. Enter task description.
4. Select a due date.
5. Save the task.

Expected Result: The new task should immediately appear in the user's task list.

Test Case 5 – Edit Task

Test Case ID: TC-Task-02
Requirement ID: TM-F-011
Objective: Verify that an existing task can be edited.
Precondition: The user has an existing task.
Test Steps:

1. Open a task.
2. Select Edit.
3. Change the title or description.
4. Save the changes.
5. Refresh the page.

Expected Result: The edited details should remain stored after the page refresh.

Test Case 6 – Update Task Status

Test Case ID: TC-Task-04
Requirement ID: TM-F-013
Objective: Verify that task status can be updated.
Precondition: A task already exists.
Test Steps:

1. Open the task.
2. Change status from To-Do to In Progress.
3. Save.
4. Change status to Completed.

Expected Result: The task status should update correctly and dashboard counts should reflect the new status.

Test Case 7 – Task Assignment

Test Case ID: TC-Collab-02
Requirement ID: TM-F-041
Objective: Verify that a task can be assigned to another team member.
Precondition: The other user is already a member of the project.
Test Steps:

1. Open a project task.
2. Select Assign.
3. Select another team member.
4. Save the task.

Expected Result: The assigned task should appear in the selected member's task list.

Test Case 8 – Performance Test

Test Case ID: TC-Perf-01
Requirement ID: TM-NF-001
Objective: Verify application response time under normal load.
Precondition: The application and test environment are available.
Test Steps:

1. Simulate normal application usage.
2. Test page loading and API calls.
3. Record response times.

4. Calculate the 90th percentile response time.

Expected Result: At least 90% of requests should complete within 2 seconds for normal load of up to 500 concurrent users.

Test Case 9 – Unauthorized Access

Test Case ID: TC-Sec-04
Requirement ID: TM-SR-004
Objective: Verify Role-Based Access Control.
Precondition: User A and User B have separate accounts.
Test Steps:

1. Login as User A.
2. Create a private task.
3. Logout.
4. Login as User B.
5. Attempt to access User A's task.

Expected Result: The request should be rejected and access should not be granted. The SRS specifies a 403 response for this condition.

Test Case 10 – Authentication Rate Limit

Test Case ID: TC-Sec-05
Requirement ID: TM-SR-005
Objective: Verify protection against excessive authentication requests.
Precondition: The authentication endpoint is available for testing.
Test Steps:

1. Send 10 authentication requests from the same IP address within one minute.
2. Send an eleventh authentication request within the same minute.

Expected Result: The eleventh request should be rejected with HTTP status 429.

13.1 Requirements Traceability Matrix

Requirement ID	Requirement	Test Case
TM-F-001	User Registration	TC-Auth-01
TM-F-002	User Login	TC-Auth-02
TM-F-003	Password Hashing	TC-Auth-03 / TC-Sec-02
TM-F-010	Create Task	TC-Task-01
TM-F-011	Edit Task	TC-Task-02
TM-F-013	Task Status Update	TC-Task-04
TM-F-041	Task Assignment	TC-Collab-02
TM-NF-001	Response Time	TC-Perf-01
TM-SR-002	Password Security	TC-Sec-02
TM-SR-004	Role-Based Access Control	TC-Sec-04
TM-SR-005	Authentication Rate Limiting	TC-Sec-05

These IDs follow the same naming used in the SRS and therefore provide proper SRS-to-test-plan traceability.

14. Test Metrics and Reporting

- Total number of test cases
- Number of passed test cases
- Number of failed test cases
- Number of blocked test cases
- Number of defects identified
- Number of defects resolved
- Functional requirement coverage
- Security requirement coverage
- Response time measurements

Test Pass Percentage

Test Pass Percentage = (Passed Test Cases / Total Executed Test Cases) × 100

A final Test Summary Report will be prepared after testing.

15. Approvals

Role	Name	Signature / Date
Team Member	R Suhaas	R Suhaas
Team Member	Rachel Sara Mathew	Rachel Sara Mathew
Team Member	Rachepalli Nandhitha	Rachepalli Nandhitha
Team Member	Rachith M	Rachith M
Faculty / Evaluator		